Vicky kumar

Curriculum Vitae

Indian Institute of Science Education and Research (IISER), Kolkata
(+91) 6206574982
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GitHub LinkedIn

Research Experience

Indian Institute of Science Education and Research (IISER) Kolkata

2024 ***Spectral Body Wave Analysis for Source Parameters Estimation.***

Conducted spectral analysis of body seismic waves using **SAC (Seismic Analysis Code)** to estimate earthquake source parameters. Performed signal processing and spectral modeling to analyze seismic wave characteristics for earthquake studies.

Advisor : **Prof. Supriyo Mitra**, *Department of Earth Sciences, IISER Kolkata*

2024 ***Seismic Data Analysis Calculator.***

Developed computational tools for seismic data analysis, including epicentral distance calculation, earthquake location estimation, and seismic hazard assessment. Designed data-driven workflows to improve earthquake analysis and disaster preparedness.

2023 ***Sedimentology Fieldwork, Himalayan Siwalik Basin.***

Applied quantitative data analysis techniques to sedimentary rock samples collected from the Himalayan Siwalik Basin to investigate depositional history, sedimentary processes, and basin evolution, providing insights into past environmental changes.

Advisor : **Prof. Sanjay K. Mandal**, *Department of Earth Sciences, IISER Kolkata*

Indian Institute of Management Kashipur (IIM Kashipur)

May, 2025 – Aug, 2025 ***Crop Residue-to-Industry Recommendation Model.***

Developed a recommendation model for predicting optimal industrial utilization pathways for agricultural crop residues using data-driven techniques, aiming to improve sustainable biomass management and resource utilization.

2025 **Live Project: Supply Chain Optimization Tool.**

- Developed an optimization-based decision support tool for a supply chain company with 400+ employees.
- Designed scheduling and workforce coordination system to improve task allocation efficiency.
- Implemented route optimization for field employees to minimize travel time and operational costs.
- Utilized Gurobi optimization solver for mathematical modeling and cost minimization.
- Improved overall operational efficiency through data-driven resource planning and scheduling.

Advisor : **Prof. Sunil Kumar Jauhar**, *Indian Institute of Management Kashipur*

Relevant Coursework

Probability and Statistics **Linear Algebra**

Data Structures and Algorithms **Calculus and Optimization**

AI / ML **Database Management and Warehousing**

Academic Achievements & Recognitions

2024 **International Youth Math Challenge 2024 (IYMC 2024)** (Germany), completed both the Qualification Round and Pre-Final Round.

2021 **Qualified JEE Main**, ranked in the top 0.1% of candidates.

2021 **Qualified IISER Aptitude Test (IAT)**, ranked in the top 0.06% of candidates.

Computer skills

Programming Languages	Python, PyTorch, R, C, C++,
Web Technologies	HTML 5, PHP, JSP, Javascript
Database	SQL, MySQL, Apache
Tools	Gurobi, Git, LATEX, Numpy/Scipy, Pytorch,

Position of Responsibility

2023 **Event Management & Logistics**, *cult fest -IISER Kolkata.*

2022 **Organizer, Team Member: E-Cell**, *IISER Kolkata.*
- Sponsored Team (Inquivesta - Sci Fest)

2022 **Hostel Representative**, *, IISER Kolkata.*

Education

2021 – 2025 **BS in Earth Science**, *Indian Institute of Science Education and Research (IISER) Kolkata,* Interdisciplinary science - climate data science & analytics

2020 **Class 12 : 83.8%**, *College of Commerce, Arts & Science, Patna.*

2018 **Class 10 : 82.8%**, *Himalayan Residential School, Patna.*

Referees

Dr. Sunil Kumar Jauhar
Assistant Professor,
Operations Management & Decision Sciences
IIM KASHIPUR
✉ sunil.jauhar@iimkashipur.ac.in